Deployment

> Part of the Streamlit curriculum. This document covers Streamlit Community Cloud, containerized deployment, and secrets management.

1. Streamlit Community Cloud

The path of least resistance for a Streamlit app is Streamlit Community Cloud — connect a GitHub repository containing the app, point it at the entry-point script, and the platform builds and hosts it for you (free for public repositories). This matches Streamlit's whole value proposition (§1 of Core Concepts): the fastest possible path from working code to a shareable, running app, with essentially no deployment engineering required.

2. Containerized deployment

For deployment outside Streamlit Community Cloud (an internal server, a different cloud platform), a Dockerfile is straightforward, since a Streamlit app is just a Python process listening on a port:

```
FROM python:3.12-slim

WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY . .

EXPOSE 8501
CMD ["streamlit", "run", "app.py", "--server.port=8501", "--server.address=0.0.0.0"]
```

```
docker build -t my-dashboard .
docker run -p 8501:8501 my-dashboard
```

Once containerized, the same options discussed for a Python API (Python API Backend, Deployment, §5) apply here too — a managed container platform (Cloud Run, ECS/Fargate) is usually the simplest way to run it continuously without managing a server yourself.

3. Secrets management

Configuration values that shouldn't be committed to source control (API keys, database credentials) are handled via `st.secrets`, backed by a `.streamlit/secrets.toml` file (kept out of version control) locally, or the hosting platform's own secrets UI in production:

```toml
# .streamlit/secrets.toml (never committed)
[database]
url = "postgresql://..."
```

```python
db_url = st.secrets["database"]["url"]
```

This is the Streamlit-specific mechanism for the same "configuration over hardcoding" principle discussed throughout this series — see the architecture overview, §2.7.

4. Core tools summary

Tool	Used for
Streamlit Community Cloud	zero-infrastructure hosting directly from a GitHub repository
Docker	packaging the app for deployment anywhere else
`st.secrets`	managing credentials/configuration without committing them to source control

See also

- Streamlit index
- Testing — verifying the app before it's deployed
- Architecture Overview, §11